

AI-BASED HUMAN PRESENCE DETECTION THROUGH WIFI SIGNAL DISTORTION

IoT - Connecting the World

**Ability Enhancement Course (AEC)
MVJ22A4071**

LEVEL II

Submitted in partial fulfillment of the requirement for the award of the degree

Bachelor of Engineering

In

DEPARTMENT OF IIOT

Submitted By

ABHILASH V	1MJ24AI002
AKSHAY MALLYA	1MJ24AI006
ANANYA M	1MJ24AI007
C NAYAN KARTHIK	1MJ24CD010
GRACIA DAFROSE	1MJ24AI021

Under the Guidance of

Dr. Manu Raju

Assistant Professor

Department of Electronics and Communication Engineering, MVJCE



**MVJ College of Engineering, Bengaluru
(An Autonomous Institute)**

Affiliated to VTU, Belagavi, Approved by AICTE, New Delhi
Recognized by UGC with 12(f) & 12(b), Accredited by NBA & NAAC

JUNE 2026



**MVJ COLLEGE OF ENGINEERING,
BENGALURU-560067**

(Autonomous Institution Affiliated to VTU, Belagavi)

DEPARTMENT OF IIOT

CERTIFICATE

Certified that the AEC Level II work titled **AI-BASED HUMAN PRESENCE DETECTION THROUGH WIFI SIGNAL DISTORTION** is carried out by the students of MVJ College of Engineering, Bengaluru, in partial fulfillment of the requirements for the award of Bachelor of Engineering during the academic year 2025–2026.

Signature of Guide

Dr. Manu Raju
Assistant Professor
Dept. of ECE

Signature of Head of Department

Dr. Thilagaraj M
Head of Department
Dept. of IIOT

External Viva

Name of Examiners

Signature with Date

1.

2.



**MVJ COLLEGE OF ENGINEERING,
BENGALURU-560067**

(Autonomous Institution Affiliated to VTU, Belagavi)

DEPARTMENT OF IIOT

DECLARATION

We hereby declare that the AEC Level II titled **AI-BASED HUMAN PRESENCE DETECTION THROUGH WIFI SIGNAL DISTORTION** has been carried out by us and submitted in partial fulfillment of the requirements for the award of Bachelor of Engineering during the academic year 2025–2026.

Further, we declare that this work has not been submitted previously for the award of any Degree or Diploma to any University.

Place: Bengaluru

Date:

Name

Signature

- | | |
|--------------------|--------------|
| 1. ABHILASH V | (1MJ24AI002) |
| 2. AKSHAY MALLYA | (1MJ24AI006) |
| 3. ANANYA M | (1MJ24AI007) |
| 4. C NAYAN KARTHIK | (1MJ24CD010) |
| 5. GRACIA DAFROSE | (1MJ24AI021) |

ACKNOWLEDGEMENT

We express our sincere gratitude to our beloved Principal, Dr. Ajayan K R, for his support towards this project work.

We express our heartfelt gratitude to **Dr. M Brindha**, Dean, School of Electrical Science, for her constant encouragement, valuable guidance and support throughout the course of our project.

Our sincere thanks to **Dr. Thilagaraj M**, Head of the Department, IIOT, MVJCE for his support and encouragement.

We are indebted to our guides, **Dr. Manu Raju**, Assistant Professor, for their wholehearted support, suggestions and invaluable advice throughout this project work.

We thank all the technical and non-technical staff members of MVJCE for their assistance.

Thank You

ABSTRACT

The rapid growth of the Internet of Things (IoT) and Artificial Intelligence (AI) has created opportunities for developing intelligent systems capable of understanding and interacting with their environment. Human presence detection is one such important application that plays a vital role in smart homes, security systems, healthcare monitoring, occupancy management, and energy-efficient automation. Traditional presence detection systems commonly rely on cameras, infrared sensors, wearable devices, or motion detectors. While these methods have been widely adopted, they often suffer from several limitations such as privacy concerns, high installation costs, dependence on lighting conditions, and the requirement for additional hardware infrastructure.

This project presents an AI-Based Human Presence Detection System that utilizes WiFi signal distortion to identify the presence of a person in an indoor environment. Whenever a human body enters an area covered by WiFi signals, the wireless signals experience variations due to reflection, scattering, absorption, and diffraction. These variations create unique patterns that can be analyzed using machine learning algorithms. By collecting WiFi signal information through an ESP32 module and processing the data using AI techniques, the system can determine whether a person is present or absent within a monitored area.

The proposed solution offers a privacy-preserving alternative to camera-based surveillance because it does not capture images, videos, or personal information. Instead, it relies solely on changes in wireless signals that are already available in most indoor environments. The system is low-cost, easy to deploy, and capable of operating in real-time. Experimental results demonstrate that the machine learning model can accurately classify human presence with an accuracy of approximately 93 percent, making it suitable for practical applications such as intrusion detection, smart home automation, occupancy monitoring, and healthcare assistance.

TABLE OF CONTENTS

CERTIFICATE	1
DECLARATION	2
ACKNOWLEDGEMENT	3
ABSTRACT	4
CHAPTER 1 INTRODUCTION	1
CHAPTER 2 LITERATURE SURVEY	3
CHAPTER 3 PESTEL ANALYSIS	6
CHAPTER 4 DESIGN ANALYSIS	9
CHAPTER 5 INDUSTRIAL DESIGN	12
CHAPTER 6 SYSTEM DESIGN AND ARCHITECTURE	14
CHAPTER 7 CONCEPTUALIZATION	16
CHAPTER 8 HARDWARE AND SOFTWARE REQUIREMENTS	18
CHAPTER 9 RESULTS AND ANALYSIS	21
CHAPTER 10 CONCLUSION	24
REFERENCES	26

CHAPTER 1

INTRODUCTION

In recent years, the demand for intelligent monitoring systems has increased significantly due to the rapid expansion of smart environments. Smart homes, smart offices, industrial facilities, healthcare centers, and public infrastructures all require efficient methods for detecting and monitoring human presence. Accurate presence detection enables automated decision-making, improved security, efficient resource utilization, and enhanced user experiences.

Conventional human presence detection systems typically employ technologies such as CCTV cameras, Passive Infrared (PIR) sensors, ultrasonic sensors, and wearable devices. Although these technologies provide effective monitoring capabilities, they possess several drawbacks. Camera-based systems raise privacy concerns because they continuously capture visual information. PIR sensors are limited to detecting motion and often fail when a person remains stationary for an extended period. Wearable devices require active participation from users and are not suitable for continuous monitoring. Furthermore, many of these solutions involve additional installation and maintenance costs.

Wireless communication technologies offer a promising alternative for human presence sensing. WiFi signals are already available in most indoor environments through routers, access points, and connected devices. When a human body interacts with these wireless signals, measurable distortions occur in signal characteristics such as Received Signal Strength Indicator (RSSI) and Channel State Information (CSI). These signal distortions contain valuable information that can be used to determine whether a person is present within the environment.

The proposed project leverages these WiFi signal distortions to create a non-intrusive human presence detection system. An ESP32 module continuously monitors WiFi signals and records variations caused by human presence. The collected data is then processed using machine learning algorithms that learn to distinguish between signal patterns associated with occupied and unoccupied environments. By combining WiFi sensing with AI techniques, the system provides a reliable and privacy-friendly solution for real-time presence detection.

The use of WiFi signals for sensing applications has gained considerable attention in recent years due to advancements in wireless communication and machine learning technologies. Unlike traditional sensing methods, WiFi-based sensing does not require users to carry any devices or remain within the field of view of a camera. This makes it an attractive solution for applications where privacy, convenience, and affordability are important considerations.

CHAPTER 2

LITERATURE SURVEY

1. Room-Level Occupancy Estimation via Wi-Fi Signal Analysis Using Low-Cost ESP32 Nodes

2026

Authors: S. Chaudhari et al.

Summary: This study proposed a room-level occupancy detection system using low-cost ESP32 devices and Wi-Fi signal analysis. Machine learning techniques were used to identify changes in wireless signal patterns caused by human presence, achieving accurate occupancy estimation for smart building applications.

2. Wi-Fi Sensing for Motion Detection using ESP-32 and Raspberry Pi 4: A Comparison

2025

Authors: Ibrahim Aldarmaki et al.

Summary: This work compares ESP32 and Raspberry Pi 4 for Wi-Fi sensing applications. CSI data was collected and analyzed using CNN and LSTM models. The ESP32-based system achieved an accuracy of 99.26 percent and demonstrated reliable human motion detection.

3. Real-Time Human Activity Detection Using Wi-Fi Signals and ESP32

2025

Authors: R. R. Kumar et al.

Summary: This research developed a real-time human activity detection framework using Wi-Fi signal variations collected by ESP32 modules. The system employed signal preprocessing and machine learning algorithms to recognize human activities accurately while maintaining privacy and low deployment costs.

4. RoFi: Robust Wi-Fi Intrusion Detection via Distribution Matching

2024

Authors: Xu Wang et al.

Summary: The authors developed an intrusion detection framework based on CSI auto-correlation modeling and likelihood ratio testing. The system achieved 97.5 percent accuracy without requiring environment-specific calibration.

5. WiFi Sensing for Presence Detection Using Deep Learning

2024

Authors: Aysha Alteneiji et al.

Summary: This work explored human presence detection using Wi-Fi signal distortions and deep learning models such as CNNs and LSTMs. The proposed system achieved high detection accuracy in both Line-of-Sight (LoS) and Non-Line-of-Sight (NLoS) environments, demonstrating the potential of AI-driven Wi-Fi sensing for reliable indoor monitoring.

6. Real-Time Object Detection for Wi-Fi CSI-Based Multiple Human Activity

Recognition

2023

Authors: Israel Elujide et al.

Summary: The study converts CSI measurements into time-frequency images and applies Mask R-CNN for activity recognition. The system achieved an average accuracy of 93.8 percent while supporting multiple activity classifications.

7. Lightweight Passive Human Motion Detection with Wi-Fi

2021

Authors: Xu Wang et al.

Summary: A lightweight motion detection approach based on selective CSI subcarriers was proposed. The system improved computational efficiency while maintaining high detection accuracy.

8. Indoor Through-the-Wall Passive Human Target Detection with Wi-Fi

2019

Authors: Aysha Alteneiji et al.

Summary: This work explored human presence detection using Wi-Fi signal distortions and deep learning models such as CNNs and LSTMs. The proposed system achieved high detection accuracy in both Line-of-Sight (LoS) and Non-Line-of-Sight (NLoS) environments,

demonstrating the potential of AI-driven Wi-Fi sensing for reliable indoor monitoring.

9. Whole-Home Level Intrusion Detection System using Wi-Fi Enabled IoT

2018

Authors: Yue Jin et al.

Summary: The proposed system utilized RSSI measurements from IoT devices for large-scale home intrusion detection and achieved near-perfect detection performance.

10. Robust Passive Static Human Detection with Commodity Wi-Fi Devices

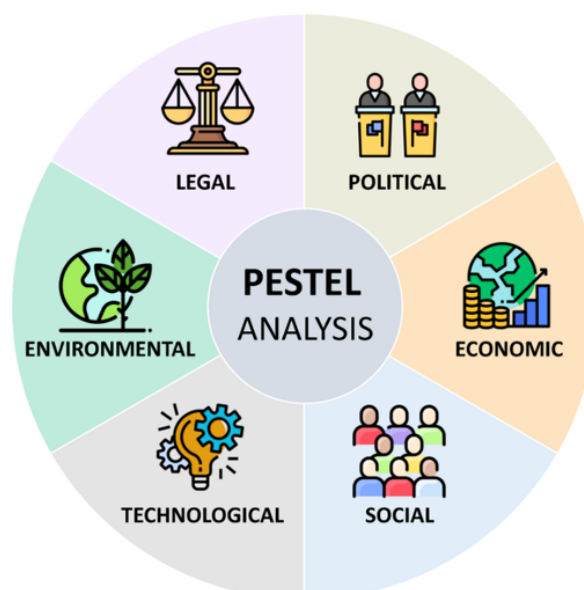
2017

Authors: Hai Zhu et al. Summary: The researchers used CSI phase information to identify static human presence through respiration-based signal variations, achieving high detection accuracy without requiring wearable devices.

CHAPTER 3

PESTEL ANALYSIS

- PESTEL Analysis is a strategic framework used to identify and evaluate the external factors that can influence the development, implementation, and success of a project. PESTEL stands for Political, Economic, Social, Technological, Environmental, and Legal factors. It helps in understanding opportunities, challenges, and risks associated with the project environment.
- For the AI-Based Human Presence Detection through Wi-Fi Signal Distortion system, PESTEL analysis provides insight into how external factors affect the adoption and performance of Wi-Fi sensing technologies in smart environments.



3.1 POLITICAL

1. Smart City Initiatives: Governments worldwide are promoting smart city projects that encourage the use of IoT and AI-based monitoring systems.
2. Digital Transformation Policies: Many countries support digital infrastructure development, creating opportunities for intelligent sensing solutions.
3. Research and Innovation Funding: Government grants and funding programs for Artificial Intelligence and IoT research can accelerate the development of Wi-Fi sensing technologies
4. Public Safety Programs: Authorities encourage the deployment of intelligent monitoring systems to improve safety and security in public spaces.

3.2 ECONOMIC

1. Low Implementation Cost: The system utilizes existing Wi-Fi infrastructure, reducing installation and operational costs
2. Affordable Hardware: Low-cost devices such as ESP32 modules make the solution economically feasible for large-scale deployment.
3. Energy Cost Reduction: Occupancy-based automation can significantly reduce electricity consumption in buildings.
4. Reduced Maintenance Expenses: Since the system operates without cameras or complex sensors, maintenance requirements remain minimal.

3.3 SOCIAL

1. Privacy Preservation: Unlike CCTV-based systems, Wi-Fi sensing does not capture images or personal information, improving user acceptance.
2. Growing Smart Home Adoption: Increasing use of smart homes and intelligent buildings drives demand for occupancy detection technologies.
3. Healthcare and Elderly Monitoring: The system can support healthcare applications by monitoring occupancy without intrusive devices.

4. Improved User Convenience: Automated control of appliances and security systems enhances comfort and user experience.

3.4 TECHNOLOGICAL

1. Advancements in Artificial Intelligence: Modern machine learning algorithms improve the accuracy of human presence detection.
2. Development of Wi-Fi Sensing Technologies: Improvements in CSI and RSSI analysis enable reliable occupancy detection using wireless signals.
3. Availability of IoT Platforms: Devices such as ESP32 provide cost-effective and efficient platforms for data collection and processing. Real-Time Data Processing: Enhanced computational capabilities support instant detection and monitoring of human presence.

3.5 ENVIRONMENTAL

1. Energy-Efficient Building Management: The system enables automatic control of lighting, cooling, and heating systems based on occupancy.
2. Reduced Power Consumption: Smart automation helps minimize unnecessary energy usage.
3. Minimal Additional Hardware: Utilizing existing Wi-Fi networks reduces electronic waste and hardware requirements.
4. Improved Sustainable Smart Infrastructure: The technology contributes to environmentally friendly building operations

3.6 LEGAL

1. Data Privacy Compliance: Any stored occupancy data must comply with applicable privacy and data protection regulations
2. Cybersecurity Requirements: Secure communication and data storage mechanisms are necessary to prevent unauthorized access.
3. Wireless Communication Standards: The system must comply with national and international Wi-Fi communication standards
4. Ethical Use of Monitoring Systems: Presence detection systems should be deployed responsibly while respecting user privacy and consent.

CHAPTER 4

DESIGN ANALYSIS

The proposed AI-Based Human Presence Detection System is designed to detect the presence of a human by analyzing distortions in WiFi signals. The system follows a modular architecture consisting of WiFi signal acquisition, data preprocessing, feature extraction, machine learning-based classification, and real-time notification modules. The design emphasizes low cost, privacy preservation, scalability, and real-time operation.

The system utilizes an ESP32 microcontroller as the primary sensing device. Unlike conventional presence detection systems that depend on cameras or infrared sensors, the proposed design leverages existing WiFi infrastructure. Human bodies naturally interact with electromagnetic waves by reflecting, absorbing, and scattering wireless signals. These interactions create measurable variations in Received Signal Strength Indicator (RSSI) values, which are used as indicators of human presence.

The data acquisition module continuously collects RSSI measurements from nearby WiFi transmissions. Since raw signal measurements often contain fluctuations caused by environmental noise, a preprocessing stage is incorporated into the design. This stage filters unwanted variations and normalizes the collected data to improve consistency and reliability. Proper preprocessing ensures that the machine learning model receives clean and meaningful input data.

The feature extraction module transforms the collected RSSI samples into statistical features that represent the characteristics of signal behavior. Features such as mean, variance, standard deviation, minimum value, and maximum value are extracted from the collected measurements. These features reduce data complexity while preserving important information related to occupancy detection.

A machine learning classifier forms the core of the system. The model is trained using datasets

collected under both occupied and unoccupied conditions. During training, the classifier learns to recognize patterns associated with human presence. Once trained, the model can accurately classify incoming signal data in real time. The use of machine learning enables the system to identify complex relationships in the data that cannot be effectively captured using simple threshold-based methods.

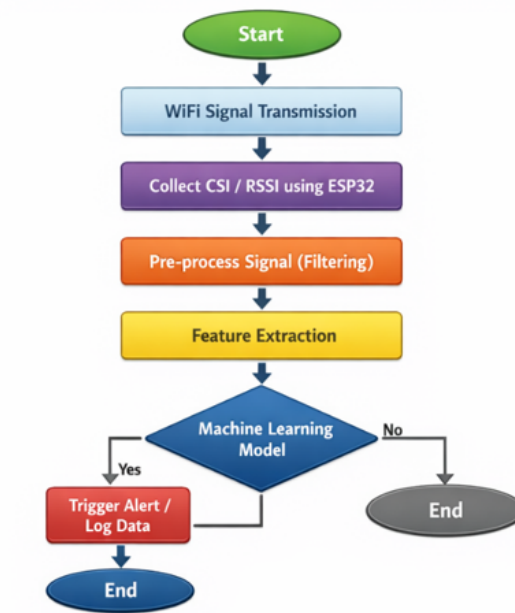
The system also incorporates a notification module through Telegram Bot integration. Whenever human presence is detected, a notification can be transmitted to the user instantly. This feature improves the practicality of the system for security monitoring, intrusion detection, and smart home automation applications.

The overall design focuses on minimizing hardware requirements while maintaining reliable detection performance. The use of a single ESP32 module and existing WiFi infrastructure significantly reduces implementation costs compared to camera-based surveillance systems. Furthermore, since the system does not capture images, audio, or personal information, it preserves user privacy and is suitable for deployment in homes, offices, healthcare facilities, and other indoor environments.

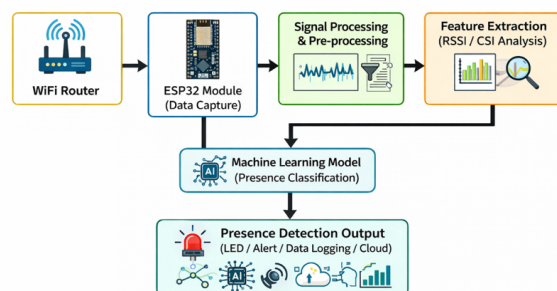
From a performance perspective, the design achieves a balance between computational efficiency and detection accuracy. The lightweight implementation allows real-time processing while maintaining an overall classification accuracy of approximately 93 percent. The modular nature of the design also enables future enhancements such as activity recognition, multi-person detection, cloud connectivity, and edge AI deployment.

Overall, the proposed system design demonstrates an effective integration of WiFi sensing, IoT technology, and machine learning techniques to create a reliable, scalable, and privacy-friendly human presence detection solution.

Flowchart of Human Presence Detection



AI-Based Human Presence Detection Using WiFi



CHAPTER 5

INDUSTRIAL DESIGN

The industrial design of the proposed AI-Based Human Presence Detection System focuses on developing a practical, low-cost, and scalable solution suitable for real-world deployment. The system is designed to operate using existing WiFi infrastructure, reducing installation complexity and minimizing additional hardware requirements. This makes the solution economically viable for large-scale adoption in residential, commercial, and industrial environments.

From a manufacturing perspective, the system requires only a small number of readily available components, primarily an ESP32 development board and a WiFi router or hotspot. The compact size of the ESP32 allows the sensing unit to be easily integrated into existing electronic devices, wall-mounted enclosures, or smart home systems. The low power consumption of the ESP32 also makes the system suitable for continuous operation without significant energy costs.

The proposed design emphasizes user privacy and safety. Unlike camera-based surveillance systems, the system does not capture images, videos, or personal information. Instead, it analyzes variations in WiFi signals, making it suitable for deployment in locations where privacy is a critical concern, such as homes, offices, hospitals, classrooms, and elderly care facilities.

The system is designed with scalability in mind. Multiple sensing units can be deployed across different rooms or buildings and integrated into a centralized monitoring system. Since the design utilizes standard WiFi communication, expansion can be achieved without major infrastructure modifications. This flexibility makes the system suitable for industrial

applications such as occupancy monitoring, security management, workspace optimization, and smart building automation.

Reliability is another important consideration in the design. The use of machine learning improves detection accuracy by adapting to environmental signal variations. The system can operate continuously and provide real-time monitoring without requiring direct user interaction. Integration with Telegram notifications further enhances usability by enabling remote access to detection results.

From an economic standpoint, the system offers significant advantages compared to traditional surveillance and sensing solutions. The elimination of specialized sensors, cameras, and wearable devices reduces procurement and maintenance costs. Additionally, the use of open-source software tools and widely available hardware components further improves affordability.

Overall, the industrial design of the proposed system provides a compact, cost-effective, privacy-preserving, and scalable solution for human presence detection. The design is suitable for deployment in smart homes, healthcare facilities, office buildings, educational institutions, and industrial environments, while also supporting future expansion and technological enhancements.

CHAPTER 6

SYSTEM DESIGN AND ARCHITECTURE

The overall architecture of the proposed system consists of multiple interconnected components responsible for data collection, processing, classification, and notification. The architecture is designed to ensure efficient operation while maintaining simplicity and scalability.

The first component is the WiFi signal source, which may be a wireless router, access point, or mobile hotspot. This source continuously transmits wireless signals throughout the monitored environment.

The second component is the ESP32 module, which acts as the signal receiver and data collection unit. The ESP32 continuously monitors WiFi signal strength and records RSSI values affected by environmental changes. Whenever a person enters or leaves the monitored area, signal characteristics change due to interaction with the human body.

The collected data is then transferred to a processing system where preprocessing operations are performed. Noise removal, normalization, and feature extraction techniques are applied to improve data quality and prepare it for machine learning analysis.

The processed features are supplied to a trained machine learning model. The model analyzes signal patterns and predicts whether human presence exists within the monitored environment. Finally, the system generates detection results and can optionally transmit alerts through a Telegram Bot. This enables remote monitoring and real-time notifications whenever human presence is detected.

The architecture ensures smooth integration between WiFi sensing, data processing, machine learning, and IoT communication technologies.

CHAPTER 7

CONCEPTUALIZATION

The conceptualization of the proposed AI-Based Human Presence Detection System is based on the principle that human presence affects the propagation of WiFi signals within an indoor environment. When a person enters the coverage area of a WiFi network, the transmitted wireless signals undergo changes due to reflection, absorption, scattering, and diffraction caused by the human body. These changes create distinct variations in signal strength, which can be analyzed to determine whether a person is present.

The project aims to utilize these naturally occurring signal distortions as a sensing mechanism for occupancy detection. Instead of relying on cameras, infrared sensors, or wearable devices, the system leverages existing WiFi infrastructure and artificial intelligence techniques to perform human presence detection in a non-intrusive manner.

The conceptual framework consists of five major stages. The first stage involves the collection of WiFi signal data using an ESP32 module. The ESP32 continuously monitors RSSI values from a WiFi source and records the signal variations occurring within the environment. The second stage involves preprocessing the collected data to remove noise and improve signal stability. This ensures that the subsequent analysis is based on meaningful information rather than random fluctuations.

In the third stage, relevant statistical features are extracted from the processed signal data. These features capture the characteristics of the signal variations and serve as inputs to the machine learning model. The fourth stage involves training a classification model using datasets collected under both human-present and human-absent conditions. The model learns to identify patterns associated with occupancy and uses this knowledge to perform predictions.

The final stage is real-time presence detection and notification. During operation, the trained model continuously analyzes incoming signal data and determines whether a person is present within the monitored area. Upon detection, the system can generate alerts and send notifications through a Telegram Bot, enabling remote monitoring and security applications. The concept focuses on creating a privacy-friendly, low-cost, and scalable human presence detection solution that can be easily integrated into smart homes, offices, healthcare facilities, and industrial environments. By combining WiFi sensing, IoT technology, and machine learning, the proposed system demonstrates an innovative approach to occupancy monitoring without requiring specialized sensing hardware or visual surveillance.

CHAPTER 8

HARDWARE AND SOFTWARE REQUIREMENTS

8.1 HARDWARE COMPONENTS

The hardware setup of the project is simple and cost-effective. It mainly consists of the following components:

1. ESP32 Development Board:

The ESP32 is the primary hardware component used in this project. It is a low-cost microcontroller with built-in WiFi and Bluetooth capabilities. The ESP32 is responsible for:

- Connecting to the WiFi network.
- Collecting RSSI values from WiFi signals.
- Monitoring signal fluctuations caused by human presence.
- Transmitting collected data for processing and analysis.
- Supporting real-time detection operations.

2. WiFi Router / Mobile Hotspot:

A WiFi router or mobile hotspot acts as the source of wireless signals used for sensing. Human presence affects the propagation of these signals, creating measurable distortions.

Functions include:

- Providing continuous WiFi signal transmission.
- Creating the sensing environment.
- Enabling communication between devices.
- Supplying signal data for analysis.

3. Computer or Laptop:

A computer system is required during the development and testing phase of the project.

Its functions include:

- Programming the ESP32.
- Collecting and storing datasets.
- Running Python-based machine learning programs.
- Training and evaluating the classification model.
- Monitoring system performance.

8.2 SOFTWARE COMPONENTS

The software components used in this project are responsible for data collection, machine learning model development, and real-time notification.

1. Arduino IDE

Arduino IDE is used for programming the ESP32 microcontroller and collecting WiFi signal data.

Functions include :

- Uploading code to the ESP32.
- Monitoring RSSI values through the serial monitor.
- Managing WiFi communication.

2. LightGBM

LightGBM is the machine learning algorithm used for classifying human presence and absence.

Advantages include:

- Fast training speed.
- High prediction accuracy.
- Efficient real-time prediction.

3. Telegram Bot API

Telegram Bot integration enables remote monitoring and notification services.

Functions include:

- Sending presence detection alerts.
- Providing real-time monitoring.
- Supporting intrusion detection applications.

CHAPTER 9

RESULTS AND ANALYSIS

The developed system was evaluated through dataset collection, machine learning model training, live prediction, and real-time notification testing. The obtained results demonstrate the effectiveness of WiFi signal distortion as a means of detecting human presence in indoor environments.

The machine learning model was trained using RSSI data collected by the ESP32 under both human-present and human-absent conditions. During the training phase, the LightGBM classifier processed a dataset consisting of more than 15,000 samples and extracted meaningful patterns from the signal variations. The training output indicated successful model convergence and generated a classification accuracy of approximately 93.07 percent, showing that the selected features were effective in distinguishing between occupied and unoccupied environments.

The achieved accuracy indicates that human presence creates measurable disturbances in WiFi signals that can be reliably identified using machine learning techniques. The model was able to correctly classify the majority of test samples, demonstrating strong learning capability and good generalization performance.

Following model training, real-time testing was performed using the live prediction module. The ESP32 continuously transmitted RSSI measurements, which were analyzed by the trained model. The system successfully identified human presence and displayed the prediction result as "HUMAN" in the terminal window. Multiple consecutive detections produced consistent outputs, confirming the stability of the prediction process and the reliability of the trained classifier during real-time operation.

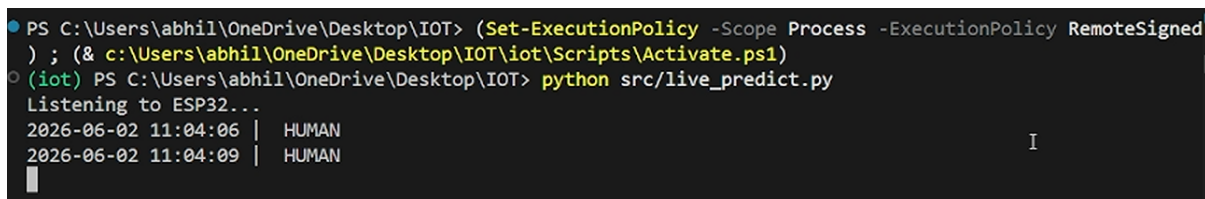
To further enhance system functionality, a Telegram Bot was integrated with the detection

module. Whenever the machine learning model detected human presence, an automatic notification was sent to the registered Telegram account. The notification included the detection status and timestamp, allowing users to monitor occupancy remotely. Successful delivery of Telegram alerts verified the proper integration between the detection system and the remote notification service.

The experimental results demonstrate that the proposed system is capable of performing real-time human presence detection using only WiFi signal information and low-cost hardware. The combination of ESP32-based sensing, machine learning classification, and Telegram notification provides a practical solution for smart home automation, security monitoring, office occupancy detection, and elderly care applications.

Although minor fluctuations in signal strength were observed due to environmental conditions and wireless interference, the trained model maintained stable performance throughout the testing phase. The overall results validate the effectiveness of the proposed approach and confirm that AI-based analysis of WiFi signal distortion can serve as a reliable, privacy-preserving, and cost-effective alternative to conventional human presence detection systems.

RESULTS:



```
PS C:\Users\abhil\OneDrive\Desktop\IOT> (Set-ExecutionPolicy -Scope Process -ExecutionPolicy RemoteSigned) ; (& c:\Users\abhil\OneDrive\Desktop\IOT\iot\Scripts\Activate.ps1)
(iot) PS C:\Users\abhil\OneDrive\Desktop\IOT> python src/live_predict.py
Listening to ESP32...
2026-06-02 11:04:06 | HUMAN
2026-06-02 11:04:09 | HUMAN
```

Figure 1: Human presence detection output obtained from the trained machine learning model.

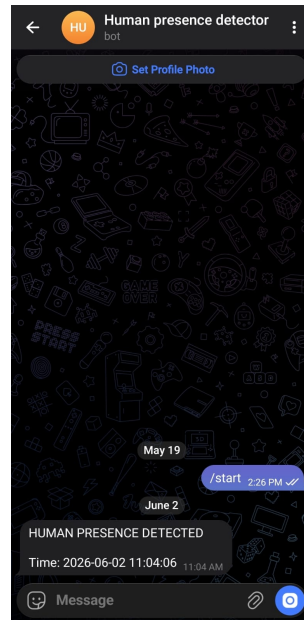


Figure 2: Telegram notification generated when human presence is detected.

```
(iot) PS C:\Users\abhil\OneDrive\Desktop\IOT> python src/main.py
And if memory is not enough, you can set `force_col_wise=true`.
[LightGBM] [Info] Total Bins 1962
[LightGBM] [Info] Number of data points in the train set: 15460, number of used features: 9
[LightGBM] [Info] [binary:BoostFromScore]: pavg=0.481759 -> initscore=-0.072995
[LightGBM] [Info] Start training from score -0.072995
C:\Users\abhil\OneDrive\Desktop\IOT\iot\Lib\site-packages\sklearn\utils\validation.py:2691: UserWarning:
X does not have valid feature names, but LGBMClassifier was fitted with feature names
  warnings.warn(
Accuracy: 93.06597671410091 %
Model saved
Done.
```

Figure 3: Accuracy Results Obtained During Model Training

CHAPTER 10

CONCLUSION

The AI-Based Human Presence Detection Through WiFi Signal Distortion project successfully demonstrates the feasibility of using WiFi signal variations as a reliable indicator of human presence in indoor environments. The system utilizes an ESP32 module to collect WiFi signal data and employs machine learning techniques to analyze changes in signal patterns caused by the presence of a person. By leveraging existing WiFi infrastructure, the proposed solution eliminates the need for cameras, PIR sensors, or wearable devices, thereby reducing deployment costs and preserving user privacy.

The collected RSSI data was preprocessed and used to train a machine learning model capable of distinguishing between occupied and unoccupied conditions. Experimental results showed that the system achieved an accuracy of approximately 93 percent, indicating that WiFi signal distortions contain sufficient information for effective human presence detection. The model demonstrated good classification performance and was able to perform real-time predictions with minimal delay.

The integration of a Telegram Bot further enhanced the functionality of the system by enabling remote monitoring and instant notifications whenever human presence was detected. This feature makes the system suitable for applications such as smart homes, security monitoring, office occupancy management, and elderly care.

Although environmental factors and wireless interference can influence system performance, the overall results confirm that WiFi-based sensing combined with artificial intelligence provides a practical, low-cost, and privacy-preserving alternative to conventional presence detection technologies. The project also highlights the potential of integrating IoT devices and

machine learning techniques to develop intelligent monitoring solutions for modern smart environments.

In conclusion, the proposed system successfully meets its objectives and demonstrates that AI-driven analysis of WiFi signal distortion can serve as an effective approach for human presence detection. With further improvements and future enhancements, the system can be extended to support advanced applications such as activity recognition, multi-person detection, healthcare monitoring, and smart building automation.

REFERENCES

1. S. Chaudhari et al., “Room-Level Occupancy Estimation via Wi-Fi Signal Analysis Using Low-Cost ESP32 Nodes,” 2026.
2. I. Aldarmaki, A. Alteneiji, K. Poon, U. Ahmed, and A. Suliman, “Wi-Fi Sensing for Motion Detection using ESP-32 and Raspberry Pi 4: A Comparison,” 2025.
3. R. R. Kumar et al., “Real-Time Human Activity Detection Using Wi-Fi Signals and ESP32,” 2025.
4. X. Wang, D. Zhang, and F. Zhan, “RoFi: Robust Wi-Fi Intrusion Detection via Distribution Matching,” 2024.
5. A. Alteneiji et al., “WiFi Sensing for Presence Detection Using Deep Learning,” 2024.
6. I. Elujide, J. Li, A. Shiran, S. Zhou, and Y. Liu, “Real-Time Object Detection for Wi-Fi CSI-Based Multiple Human Activity Recognition,” 2023.
7. X. Wang and L. Zhang, “Lightweight Passive Human Motion Detection with Wi-Fi,” 2021.
8. Y. Xiaolong, W. Shiming, and Z. Mu, “Indoor Through-the-Wall Passive Human Target Detection with Wi-Fi,” 2019.
9. Y. Jin, Z. Tian, and M. Zhou, “Whole-Home Level Intrusion Detection System using Wi-Fi Enabled IoT,” 2018.
10. H. Zhu, F. Xiao, L. Sun, X. Xie, and R. Wang, “Robust Passive Static Human Detection with Commodity Wi-Fi Devices,” 2017.